

Idea of a new approach for prompt optimization

- Optimize a set of diverse prompts instead of a single prompt
- Organize prompts into clusters capturing different aspects of the task, by splitting errors/feedback (with some overlap) across clusters.
- Perform local optimization within clusters (error analysis, mutation, crossover)
- Allow inter-cluster crossover at the end.
- Moreover, use global error analysis to guide overall improvement
- Combine outputs from multiple final prompts to make decision. Here are some ideas to optimize the inference cost:
 - Use only a subset of the prompts to reduce inference cost.
 - Start with the best prompt, and iteratively add others until we are confident in the prediction.
 - Use a router or classifier to map inputs to the most suitable prompt(s).

Other updates I evaluated several optimized prompts (from earlier experiments) on the FS-TACRED test set, and they show improvement over the baseline prompt. However, prompts optimized without feedback show a drop in performance, because they were optimized directly on the validation set (overfit), whereas other approaches obtain feedback from the training set.